

Fig. 3. Evoke System Architecture (10 AWS Services & 3 External APIs)

Identity & Access Control Boundary — Amazon Cognito + API Gateway (Enforces Invariant I1)

INGESTION & EXTRACTION

CONVERSATION LAYER

CLIENT & OBSERVABILITY

Amazon S3

Encrypted Vault (AES-256)

Amazon Transcribe

Speaker Diarization ASR

Amazon Comprehend

Multi-dim NLP (Keys K)

Schema-Builder Lambda

PIS Population & Versioning

Amazon DynamoDB

Schema Store (TTL I2)

Amazon API Gateway

Authenticated REST Endpoint

Orchestration Lambda

Humility Routing Gate (τ)

Groq API (Llama-3-70B)

Primary Inference (<800ms)

Gemini 1.5 Flash

Automatic Model Failover

ElevenLabs API

Neural Voice Synthesis

AWS Amplify

Zero-Config Web Hosting

React / Next.js Client

Quiet Converse UI & Vault

Amazon Polly

Dev/Testing Speech Quota

Amazon Cognito

Role Separation (I1)

Amazon CloudWatch

Immutable Audit Logs (I4)